04 — Lifecycle Manager: end-to-end report

Date: 2026-10-25 **Two e2e surfaces:** disposable Kind (this Mac) and live dh2 (BlueField-2).

1. Kind e2e (disposable cluster)

Purpose

Prove the adapter manager, metrics, mock VSP, NVIDIA emit path, chart NF, and **Helm mutation on a dedicated release** — never live `dpf-operator`.

Commands

```
cd opi-nvidia-dpf-adapter
export GOTOOLCHAIN=go1.26.10
make docker-preflight
make test-e2e
```

`docker-preflight` must see a working Docker engine. If `unix:///Users/kartikey0104/.docker/run/docker.sock` is missing, the target **fails closed**. That is intentional.

Historical failure (in-repo evidence)

File: `opi-nvidia-dpf-adapter/evidence/e2e-tests.log`

```
failed to connect to the docker API at unix:///Users/kartikey0104/.docker/run/docker.sock
make: *** [setup-test-e2e] Error 1
```

Sandbox runs also got `permission denied` on the socket. That log is **not** the current result. It is the reason preflight exists.

Current result (after preflight)

Kind cluster name: `opi-nvidia-dpf-adapter-test-e2e` Outcome: **6/6 PASS** Cluster: **deleted** at the end of the suite (each run).

#	Spec	What it proves	Result
1	Manager runs	controller Deployment Ready	PASS
2	Metrics	metrics endpoint serving	PASS
3	Mock VSP	unix VSP stamps DataProcessingUnit serial	PASS
4	Emit DPF objects	DPUDevice, DPUFlavor, BFB, DPU from annotated DPU	PASS
5	Helm-chart NF	DPUService for chart NF; image-only NF skipped	PASS
6	HelmProbe	install 0.1.0 → upgrade 0.2.0 → rollback 0.1.0 → uninstall	PASS

HelmProbe (the LCM e2e)

File: `test/e2e/helm_probe_test.go` Identity: namespace/name `opi-lcm-probe`, chart `pkg/lifecycle/testdata/lcm-probe/{0.1.0,0.2.0}`.

Guarantees:

- Never uses release name `dpf-operator`
- Never talks to NGC
- Uninstall leaves no probe Secret/ConfigMap

- Proves `ClusterHelmClient` mutation without importing Helm SDK

Unit companion (faster than Kind):

```
go test ./pkg/lifecycle/ -count=1 -run 'TestClusterHelmClient|TestConcurrentClusterHelm'
```

Kind versions

CI pins Kind **v0.32.0** (not `latest`). Kubernetes fixtures align with the v1.36 pin.

2. Live lab user-flow e2e (dh2)

This is not Kind. This is the real BlueField-2 + DPF v26.4.0 path.

Connect

```
# VPN Mac
ssh -o IdentityAgent=none -o IdentitiesOnly=yes \
  -o IdentityFile=$HOME/.ssh/opi_lab_key -o UseKeychain=yes \
  -o KexAlgorithms=curve25519-sha256 \
  root@172.22.1.2

k3s kubectl get nodes
k3s kubectl get dpuoperatorconfig
k3s kubectl -n dpf-operator-system get pods
k3s kubectl -n dpf-operator-system get dpu
```

Expected user steps

1. Enable NVIDIA (`spec.vendor=nvidia` or annotation) 2. LCM preflight + dependency check (k8s 1.36, cert-manager, Argo) 3. Observe/claim Helm pin `dpf-operator-system/dpf-operator@v26.4.0` 4. LCM conditions Supported + VendorOperatorReady 5. DPF provisions the DPU (BFB, NVCONFIG, reboot) 6. Dataplane: HBN / flannel / CNI / SFC / SR-IOV

Last recorded live result (2026-10-25 ~03:30Z)

Step	Result
1 Enable NVIDIA	`spec.vendor=nvidia`, `adopt-existing=true`
2 Preflight	k3s 1.36.4, cert-manager, Argo, NFD, maintenance-operator present
3 Helm	Secret claimed; adapter owner + mode=adopted; storage owner=helm
4 LCM	`action=noop` `installed=v26.4.0` Supported=True VendorOperatorReady=True
5 DPF provision	**STALL** — DPU `dh2-bluefield-2` phase **Rebooting**
6 Dataplane	`DPFReady=False`; hbn / flannel / cni-installer / sfc **Pending**; sriov-device-plugin Pending ~37h

DPU annotations / reboot:

- `provisioning.dpu.nvidia.com/host-power-cycle-required=true`
- `rebootMethod` / `rebootStatus.method`: **SystemLevelReset**
- Host agent wants mlxfwreset level 3 on `/dev/mst/mt41686_pciconf0` (and .1)
- Pending NVCONFIG: `PF_SF_BAR_SIZE`, `PF_TOTAL_SF`
- BFB `bf-bundle-340` prepared/ready
- Device `mt2321xz0kvx` PCI `0000-0f-00` serial `MT2321XZ0KVX`

Not run: `mlxfwreset`, host reboot, helm-uninstall, AMD stubs.

After the LCM observer patch, the intended live signal is `VendorDPUReady=False/Rebooting` with the DPU name in the message, while `VendorOperatorReady` stays True. Image on lab at last roll was `controller:lcm-20261024b` (observer

landed after that roll — roll again if you want the condition on the live object).

Compiled LCM tests on dh2

dh2 has no `go`. Mac:

```
GOTOOLCHAIN=go1.26.10 GOOS=linux GOARCH=amd64 \
  go test -c -o /tmp/lifecycle.test ./pkg/lifecycle/
```

Copy binary + `go.mod` + `config/rbac/role.yaml` + testdata into `/tmp/lab-job-<name>/`. Run. Delete job dir.

Recorded: full package PASS after go.mod was included.

3. Conformance / mapping (related e2e)

```
go test ./test/conformance/ -count=1
go test ./pkg/mapping/ -count=1
go test ./pkg/translation/ -count=1
```

Evidence unit log (`evidence/unit-tests.log`) — coverage snapshot:

Package	Time	Cover
pkg/lifecycle	2.015s	89.2%
pkg/mapping	1.369s	80.3%
pkg/translation	7.153s	72.7%
pkg/discovery	0.775s	80.4%
pkg/vsp	4.665s	73.0%
test/conformance	11.413s	—

4. What “e2e green” does and does not mean

Green Kind e2e means: manager, VSP, NVIDIA emit, chart NF, and probe Helm install/upgrade/rollback/uninstall work on a disposable cluster.

It does not mean: the BlueField finished provision. That is step 5 on dh2 and is still **Rebooting**.

Green LCM on dh2 means: pin claimed, noop, operator CRDs/controllers healthy. Dataplane can still be Pending.

5. Re-run checklist

```
# Mac
export GOTOOLCHAIN=go1.26.10
go test ./pkg/lifecycle/ -count=1
go test -race ./pkg/lifecycle/ -count=1
make govulncheck
make docker-preflight
make test-e2e

# Lab hop
ssh ... root@172.22.1.2 \
  'k3s kubectl get dpuoperatorconfig dpu-operator-config -o yaml | head
   k3s kubectl -n dpf-operator-system get dpu dh2-bluefield-2 -o yaml | head
   k3s kubectl -n dpf-operator-system get dpuservice'
```